

AM50 LAB 7

1. INTRODUCTION

The goal of today's computer lab is severalfold:

Introductory We will carry out simulations of random walks, and verify the $L(t) = \sqrt{Dt}$ relationship outlined in class.

Advanced We will read in data for the Standard and Poor's 500 Index into MATLAB, and start to ask whether there is evidence for or against the hypothesis that the data is well described by a random walk.

Remark! If you do not get to the advanced part this week and are fascinated by stocks, do not despair!! We will cover some of this ground again next week when we study option pricing. This uses random walks applied to stocks as a central part of the mission.

2. SIMULATING RANDOM WALKS

We would like to simulate a random walk. To do this we recall the model from class. We consider a walker at $x = 0$ and flip a coin to let the walker move either to the left or to the right. This is accomplished with the following MATLAB code:

```
x=0;  
if rand(1,1)< 0.5  
x=x+1;  
else  
x=x-1;  
end
```

Please note the following about this code fragment:

- (1) The command `rand(1,1)` chooses a random number in the interval $[0, 1]$. To check this, try it! Type `rand(1,1)` in the MATLAB window several times. (You can use the up arrow key to get the command to repeat.) You will see that it outputs a bunch of random numbers between $[0, 1]$. Note that if you plot `rand(10,1)` MATLAB outputs a row vector of 10 random numbers. The command `rand(1,10)` outputs a column vector of 10 random numbers. This is not useful for our lab but it is worth noting.
- (2) Note that lines need to end in semi-colons or MATLAB prints their output. Semi colons suppress output printing. You will get use to this and grow to see its use!
- (3) The next several lines are an example of an *if-then-else* statement. If the drawn random number is smaller than $1/2$ then the walker moves to the right ($x = x + 1$); otherwise the walker moves to the left.
- (4) The final command “end” tells the if-then-else statement to end.

Now we would like to iterate this code fragment many times to find out what happens. The code below iterates this code fragment in two ways: first it runs the fragment 200 (tmax) times. This corresponds to 200 steps of the random walk. Secondly, it uses 100 different walkers, each of which is started out at the origin.

```
% SET UP
tmax=200; % total number of steps for the walkers
nparticles=100; % number of random walkers
x=zeros(nparticles,1); % start out all of the walkers at x=0

% SETUP FOR PLOTTING--BEGINNERS CAN IGNORE THIS! This just helps with the plots
xpoints=-20:1:20; % places where histogram centers the bins
axis([-20 20 0 50]); % sets the axes of the plots

for i=1:tmax % first loop: move each of the walkers tmax times

    for j=1:nparticles % in each timestep, each of the nparticle walkers moves to the left
        if rand(1,1)>0.5
            x(j)=x(j)+1;
        else
            x(j)=x(j)-1;
        end
    end

end

%PLOTTING COMMANDS--BEGINNERS IGNORE, NOT IMPORTANT!
axis manual
nn=hist(x,xpoints); % trick for making a plot
bar(xpoints,nn); % trick for making a plot
M(i) = getframe; % trick for making a movie
pause(1/10);

% ANALYSIS COMMANDS--THIS LETS US PLOT THE
width(i)=std(x)/2; % half the standard deviation of the distribution
meanx(i)=mean(x); % mean of the distribution
end
movie(M); % plays the frames again as a movie!
```

Please note the following aspects of this program.

- (1) The program simulates the trajectories of nparticles particles. This is so that we can do the random walk many times.
- (2) The program contains two *loops*. Within the loop the program repeats the specified task a specified number of times. The outer loop goes through the different timesteps. The inner loops moves each particle separately.

- (3) The plotting commands are totally confusing and I recommend that you only look into these if you want to. Otherwise just take this as given to allow you to make movies and watch!! It isn't important.

Exercise: 1

Run this computer program (saved as "randomwalk.m" in your home directory). The variable "width" saves the width of the distribution as a function of time—this is calculated as the standard deviation of the distribution in the positions of the walkers at each time. Plot the width (`plot(width,'o')`) and convince yourself that the width grows according to our famous law

$$L(t) = \sqrt{Dt}.$$

A good way to convince yourself of the $\sqrt{Dt}$ law is to use a log-log plot. Note that if we calculate

$$\log(L(t)) = \log(\sqrt{Dt}) = \frac{1}{2}\log(Dt) = \frac{1}{2}\log(t) + \frac{1}{2}\log(D).$$

Thus if we plot $\log(L)$ versus $\log(t)$ we should find a straight line with a slope of $1/2$.

We also might check that the value of the diffusion constant is what we expect: In class we argued that $D = a^2/(2\tau)$, where a is the lattice spacing and τ is the timestep. In our simulation both the lattice spacing $a = 1$ and $\tau = 1$ so that $D = 1/2$.

Exercise: 2

(A) Do a log-log plot of $L(t)$ and verify that the slope is $1/2$ in accordance with the prediction. (B) Plot the average value of the walkers, by using the command `plot(meanx,'o')`. What happens to the mean? Does this make sense? (C) Let's check that the diffusion constant is correct. First plot the width `plot(width,'o')`; *hold all*; and then plot the prediction `plot(sqrt(1:tmax)*sqrt(1/2))`

2.1. **“Biased Random Walkers”**. What happens if we change the probability of moving to the left or the right? Consider the generalized code fragment

```
if rand(1,1)>R
x(j)=x(j)+1;
else
x(j)=x(j)-1;
end
```

Exercise: 3

What happens if $R = 1$? $R = 0$? $R = 0.75$? Can you explain what is going on with these behaviors? HINT: Plot the mean as a function of time in these different cases and see what happens. Also see if there is a change in the behavior of the standard deviation. Can you explain what is going on?

Exercise: 4

Can you think of an example where a biased random walker might be an appropriate model? Discuss with your partner and your TF and brainstorm what might work.

3. *Advanced: Only to be done if you get done early!!* FUN WITH STOCK DATA

Can we apply these idea to data from Stocks? To read in data from the *S&P 500* for 50 years use the command

```
M=csvread('S_P_500.csv',1,1);
```

This reads the data from January 3 1953 to February 19,2010. The data is saved for every day that the Stock Market is open. The first column $M(:,1)$ gives the opening price; the second column gives the high $M(:,2)$; the third column gives the low and the fourth $M(:,4)$ gives the close. The fifth column gives the volume of shares traded.

For some strange reason the data is stored backwards—the early dates are at the end of the table and the end dates are at the beginning. Thus if you plot

```
plot(M(:,1),'o')
```

you will see that the data backwards—with the last date occurring first and the first date occurring last. To correct this lets type

```
plot(M(end:-1:1,1),'o')
```

This reverses the plot.

Now, the data looks to be quite exponential! This makes sense of course, you are usually told to invest in stocks (or even a bank) because of exponential growth. And even inflation is exponential. A simple first bit of analysis to do is to see how well the data is fit by the law

$$S(t) = S_0 e^{rt},$$

where $S(t)$ is the price of the stock portfolio at time t , S_0 is the initial price and r is the growth rate.

Exercise: 4

To test this theory, lets make a semi-log plot of the stock data. Type `semilogy(M(end:-1:1,1))`. You will see that the data is basically a straight line. Demonstrate that the data is pretty well fit by the law $S(t) = 20e^{rt}$, with $r = 3e - 4$.

For the last demonstration, you need to plot $S(t) = 20e^{rt}$ on top of the measurements. To do this consider the following commands

```
r=3e-4;
t=0:100:15000; %
s=20*exp(r*t);
semilogy(t,s);
```

Finally we do not have enough time left in this lab to consider this in detail but it is worth considering a little and discussing how you can test if this data obeys a

random walk. The fact that $S(t)$ increases exponentially makes it seem reasonable that the random walk will be in the rate r instead of in S itself.

Exercise: 5

(IF YOU HAVE TIME) Discuss how you might decide if the data is consistent with the random walk model.